

Concentrated Risk: Misallocation and Granular Business Cycles

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- **This paper:** Formalizes the link between misallocation and business cycles in an analytically tractable framework

- ▶ Uncovers a stability efficiency tradeoff
- ▶ Finds novel stabilization role for existing size-dependent policies

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- ❹ **Positive correlation is empirically likely**
- ❺ Propose a portable new method to evaluate business cycle implications of misallocation counterfactuals using standard models/sufficient statistics

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- ④ **Policy implications: Stabilization role for size-dependent policies (SME subsidies, EPL, etc.)**

Literature & Contribution

- ① **Granular business cycle lit.** Gabaix (2011); Acemoglu et al. (2012); Carvalho and Gabaix (2013); di Giovanni et al. (2018); Carvalho and Grassi (2019)

- ▶ Substantiating the granularity hypothesis theoretically and empirically
- ▶ Quantifying its relative importance for overall business cycle volatility

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❷ **Misallocation lit.** Restuccia and Rogerson (2008); Hsieh and Klenow (2009); Bartelsman et al. (2013); Hsieh and Klenow (2014); Hopenhayn (2014); Bento and Restuccia (2017); Poschke (2018); Buera and Fattal Jaef (2018); David and Venkateswaran (2019); Baqaee and Farhi (2020)

- ▶ Understanding differences in productivity across countries and time
- ▶ Concerned with counterfactuals and their quantification - e.g., how would a counterfactually more efficient economy compare with the observed one in terms of welfare and activity

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- 3 **This paper bridges the gap between these two literatures and**
 - ▶ Considers the granularity hypothesis in the presence of misallocation
 - ▶ Analyses the business cycle implications of commonly studied misallocation counterfactuals leveraging the granularity hypothesis + new method to conduct counterfactuals
 - ▶ Uncovers a trade-off between production efficiency and business cycle stabilization implying a novel stabilization role for size-depenedent policies (EPL, SME subsidies, etc.)

Framework - Misallocation and Granular Business Cycles

Heterogeneous Firms Model - Efficient Environment

- **Technology** - N firms that produce using a DRS technology

$$y_i = z_i l_i^\gamma, \quad \log(z_i) = \log(a_i) + \tilde{x}_i, \quad 0 < \gamma < 1 \quad (1)$$

y_i	output of i	l_i	labor input allocated to i
z_i	productive ability	$\tilde{x}_i$	stochastic productivity shifter - $E[e^{\tilde{x}_i}] = 1$ and st.dev. σ_x
a_i	deterministic ability	$\mathbf{s} \in R^N$	aggregate state - all N realizations of $\tilde{x}_i$

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- **Production** - firms maximize expected output

$$\max_{l_i} E[y_i] - w \times l_i \quad (2)$$

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$$\sum_{i=1}^N l_i = L \quad (3)$$

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- **Result** - RBC-like aggregate production function with TFP governed by granular shocks

$$Y_s = \sum_{i=1}^N y_i = \frac{\sum_{i=1}^N a_i^{\frac{1}{1-\gamma}} e^{\tilde{x}_i}}{\left[\sum_{i=1}^N a_i^{\frac{1}{1-\gamma}} \right]^\gamma} L^\gamma = \underbrace{Z_s}_{\text{TFP}} \times L^\gamma \quad (4)$$

Conceptual Validation - Known Results Hold

Hulten's Theorem (1978)

The effect of a one per cent shock to the j^{th} firm on aggregate TFP is $\eta_{j,s}$ given by

$$\eta_{j,s} = \frac{\partial \log(Z_s)}{\partial \tilde{x}_j} = \frac{a_j^{\frac{1}{1-\gamma}} e^{\tilde{x}_j}}{\sum_{i=1}^N a_i^{\frac{1}{1-\gamma}} e^{\tilde{x}_i}} = \quad (5)$$

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Business Cycle Volatility - (possible) micro-foundations for Gabaix 2011

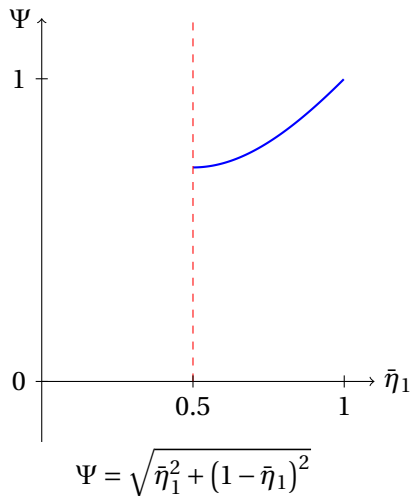
The volatility of log TFP can be approximated as

$$\sigma_Z \approx \sigma_x \sqrt{\sum_{i=1}^N \overline{\eta_i}^2} = \underbrace{\sigma_x}_{\text{micro volatility}} \times \underbrace{\Psi}_{\text{amplification term}}, \quad \overline{\eta_i} = \frac{\partial \log[Z(\mathbf{a}, \mathbf{x} = \bar{\mathbf{x}})]}{\partial \tilde{x}_i} \quad (6)$$

Building Intuition - How Misallocation Affects the Business Cycle?

Suppose that $N = 2$ and $a_1 \geq a_2$

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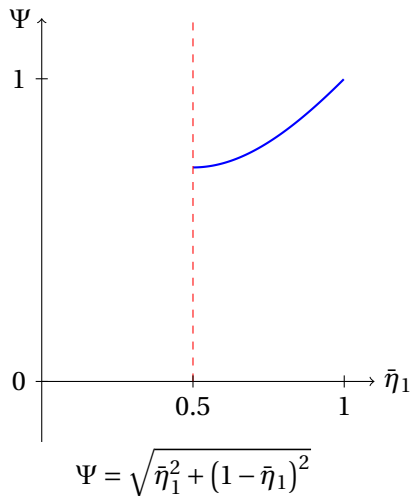


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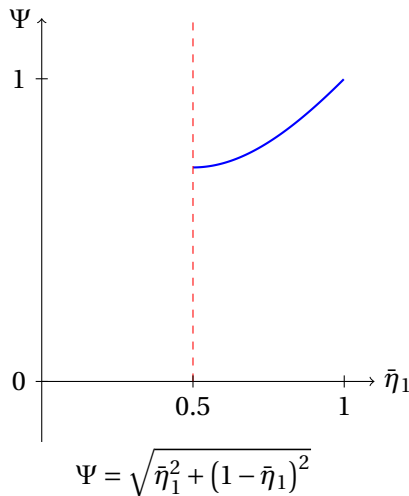
Intuition - economy with misallocation and

① Positively correlated distortions -

$$\downarrow \bar{\eta}_1 \Rightarrow \Psi \downarrow$$

② Negatively correlated distortions -

$$\uparrow \bar{\eta}_1 \Rightarrow \Psi \uparrow$$



Misallocation & Business Cycle Volatility

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- **Production** - firms maximize expected output - face implicit tax τ_i

$$\max_{l_i} E[y_i] - w l_i \quad (8)$$

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- **Result** - aggregate production is given by

$$Y_s = \sum_{i=1}^N y_i = L^\gamma \frac{\sum_{i=1}^N a_i^{\frac{1}{1-\gamma}} (1 - \tau_i)^{\frac{\gamma}{1-\gamma}} e^{\tilde{x}_i}}{\left[\sum_{i=1}^N (a_i (1 - \tau_i))^{\frac{1}{1-\gamma}} \right]^\gamma} = \underbrace{Z_s^d}_{\text{distorted TFP}} \times L^\gamma \quad (10)$$

Mapping Between the Efficient and Distorted Cases

- **Distorted** elasticities of TFP wrt a shock to firm j :

$$\delta_{j,s} = \frac{\partial \log(Z_s^d)}{\partial \tilde{x}_j} = \frac{a_j^{\frac{1}{1-\gamma}} (1 - \tau_j)^{\frac{\gamma}{1-\gamma}} e^{\tilde{x}_j}}{\sum_{i=1}^N a_i^{\frac{1}{1-\gamma}} (1 - \tau_i)^{\frac{\gamma}{1-\gamma}} e^{\tilde{x}_i}} = \underbrace{s_{Yj}^d}_{\text{sale share}} \quad (11)$$

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- **Aggregate volatility:**

$$\sigma_Z^d \approx \sigma_x \times \sqrt{\sum_{i=1}^N \overline{\delta i}^2} = \sigma_x \times \Psi^d \quad (12)$$

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- Comparing the effect of a shock to firm j in the two cases:

$$\overline{\delta_j} = \frac{a_j^{\frac{1}{1-\gamma}}}{\sum_{i=1}^N a_i^{\frac{1}{1-\gamma}}} \frac{(1 - \tau_j)^{\frac{\gamma}{1-\gamma}}}{(1 - \hat{\tau})^{\frac{\gamma}{1-\gamma}}} = \overline{\eta_j} \sqrt{1 - \textcolor{red}{d_j}}, \quad (13)$$

Theorem 1 - Misallocation and Business Cycle Volatility

$$\Delta HHI = \underbrace{\Psi^2}_{\text{Efficient benchmark}} - \underbrace{\Psi_d^2}_{\text{Distorted economy}} = \underbrace{\cos(\theta)}_{\text{'Correlation' of distortions and size}} \times \underbrace{\|\mathbf{d}\|}_{\text{Misallocation Severity}} \times \underbrace{\|\bar{\eta}^2\|}_{\text{Skewness of firm-size dist}}, \quad \cos(\theta) = \frac{\sum_{i=1}^N \bar{\eta}_i^2 d_i}{\|\mathbf{d}\| \times \|\bar{\eta}^2\|}$$

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- ② **Misallocation amplifies business cycle** if distortions are negatively correlated

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- ❶ Misallocation dampens business cycle if distortions are positively correlated
- ❷ Misallocation amplifies business cycle if distortions are negatively correlated
- ❸ 1 & 2 are **stronger** if distortions are more severe ($\|\mathbf{d}\| \uparrow$), the correlation is stronger in magnitude, or the firm-size distribution is more skewed.

Parametric Example: SME Subsidies

- 1 Consider a government that subsidizes small businesses with a subsidy that declines with the size of the firm: $(1 + s_0) \left(\frac{l_i}{L} \right)^{-\nu}$ or

$$\log(\text{Subsidy}_i) = \alpha_0 - \nu \log(\text{firm size}) \quad (14)$$

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- 5 Conclusion: **SME subsidy = ex-ante automatic stabilizer**

Discussion of Correlated Distortions

Misallocation $\Rightarrow$ MPLs are not equalized by the market mechanism

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 - 1 Corruption by big firms - ties to local gov. allows large firms to grow above optimal

Welfare Implications

- Assumptions:

- ▶ Normalize $L = 1$
- ▶ Single representative consumer with log utility (CRRA in paper) $u(C_s) = \log(C_s)$
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- ▶ **Negatively correlated distortions** - ignoring business cycle effect $\Rightarrow$ **understated welfare gains**

Quantification Strategy - The Effects of Misallocation on Volatility

Studying Counterfactuals With Their Business Cycle Implications

- ① **Ideal data requirements:** All the values of $\bar{\eta}$ and $\bar{\delta}$ - **Impossible**, even $\bar{\eta}$ are unknown sometimes
- ② **Challenge 1:** Available data usually involves only parametric description of the firm-size distribution - **Model inferred object/aggregate descriptives**
- ③ **Challenge 2:** Counterfactuals should compare an economy with N existing firms with their productive ability to an economy with the same N firms in a counterfactual scenario - **Expectations of aggregates are not enough**

Quantile-based Approximation Strategy

- 1 Exploit the quantile function Q to treat function of random sample using $a_i = Q(q_i)$

$$\Psi(\mathbf{a}) = \sqrt{\sum_{i=1}^N \overline{\eta}_i(\mathbf{a})^2} = \sqrt{\sum_{i=1}^N \left[\frac{a_i^{\frac{1}{1-\gamma}}}{\sum_{j=1}^N a_j^{\frac{1}{1-\gamma}}} \right]^2} = \sqrt{\sum_{i=1}^N \left[\frac{Q^{\frac{1}{1-\gamma}}(q_i)}{\sum_{j=1}^N Q^{\frac{1}{1-\gamma}}(q_j)} \right]^2} = \tilde{\Psi}(\mathbf{q})$$

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- 2 Suppose WLOG that the sample is ordered, in expectations - $\bar{q}_i = E[q_i] = \frac{N+1-i}{N+1}$, why?
 q_i is the i^{th} order statistic of the Uniform distribution $\Rightarrow q_i \sim \text{Beta}(N+1-i, i)$

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- 3 Take first-order Taylor series approx. around the point $\bar{\mathbf{q}}$ and take expectations:

$$E[\Psi(\mathbf{q})] \approx E\left[\Psi(\bar{\mathbf{q}}) + \sum_{i=1}^N \frac{\partial \Psi(\mathbf{q})}{\partial q_i} (q_i - \bar{q}_i)\right] = \Psi(\bar{\mathbf{q}}) \quad (16)$$

Second order - conceptually possible (not practical)

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Implication 3 - The amplification term is approximately

$$\overline{\Psi}(N, \zeta) = \frac{\sqrt{\sum_{i=1}^N i^{-\frac{2}{\zeta}}}}{\sum_{i=1}^N i^{-\frac{1}{\zeta}}} \quad (18)$$

Asymptotic Behaviour of the Amplification Term

$$\overline{\Psi}(N, \zeta) = \frac{\sqrt{\sum_{i=1}^N i^{-\frac{2}{\zeta}}}}{\sum_{i=1}^N i^{-\frac{1}{\zeta}}}, \quad (19)$$

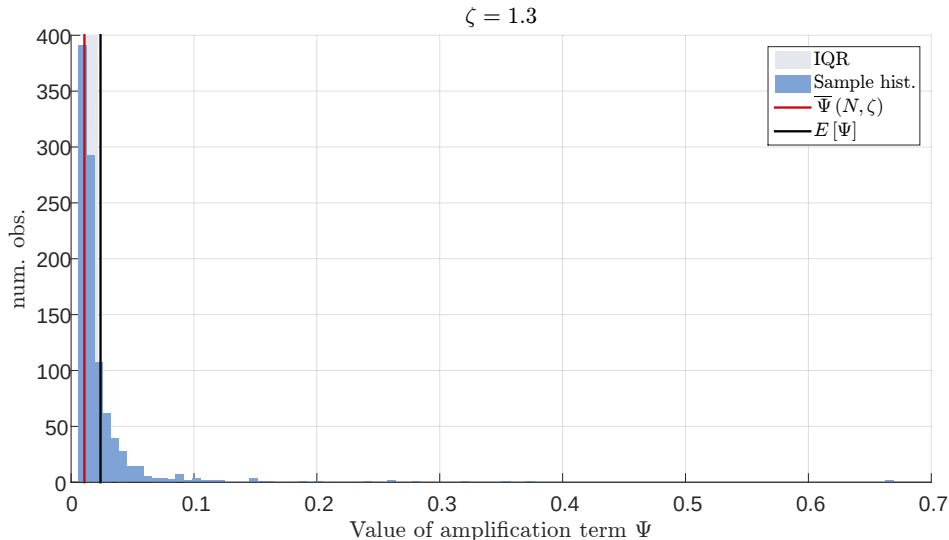
Useful fact - $\lim_{N \rightarrow \infty} \left(\sum_{i=1}^N \frac{1}{i^{\frac{1}{\zeta}}} \right) = \text{constant}, \zeta < 1 - \zeta \left(\frac{1}{\zeta} \right)$

Three cases:

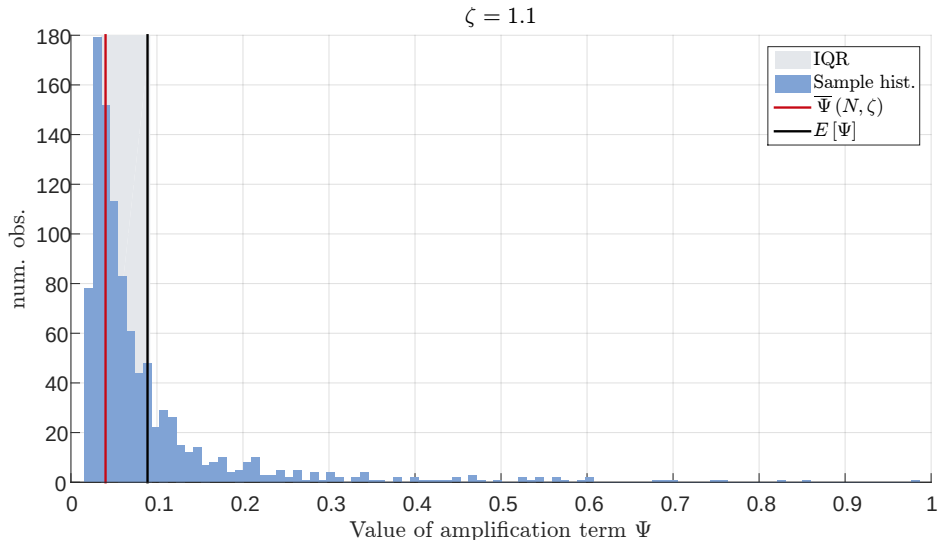
- ❶ $\zeta < 1 \Rightarrow$ No asymptotic decay - ratio of two constants (not previously shown)
- ❷ $\zeta = 1 \Rightarrow$ Decay of $\log(N)$ (Gabaix 2011)
- ❸ $1 < \zeta < 2 \Rightarrow$ Decay of $N^{1-\frac{1}{\zeta}}$ (Gabaix 2011)
- ❹ $2 < \zeta \Rightarrow$ Law of large numbers holds, Ψ behaves as $\frac{1}{\sqrt{N}}$

Validation

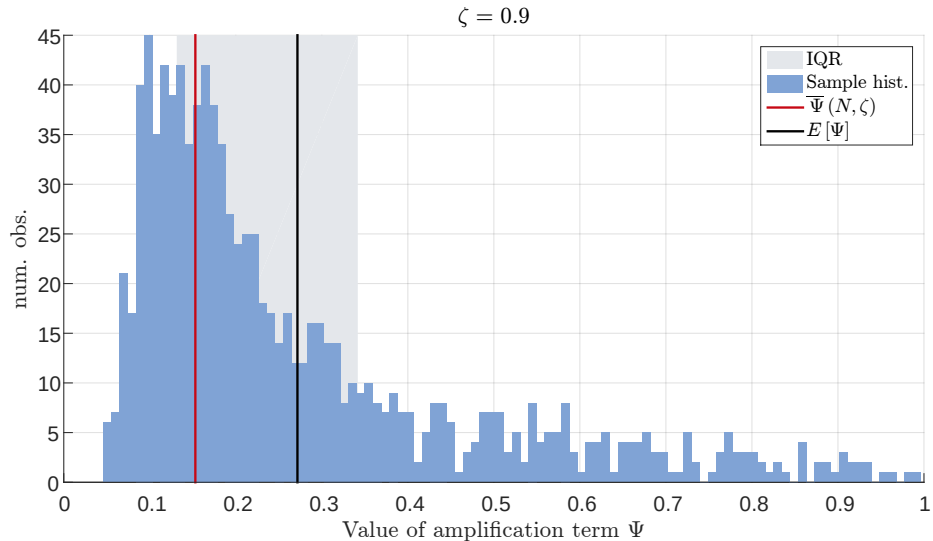
Assessment of the Approximation - Simulation



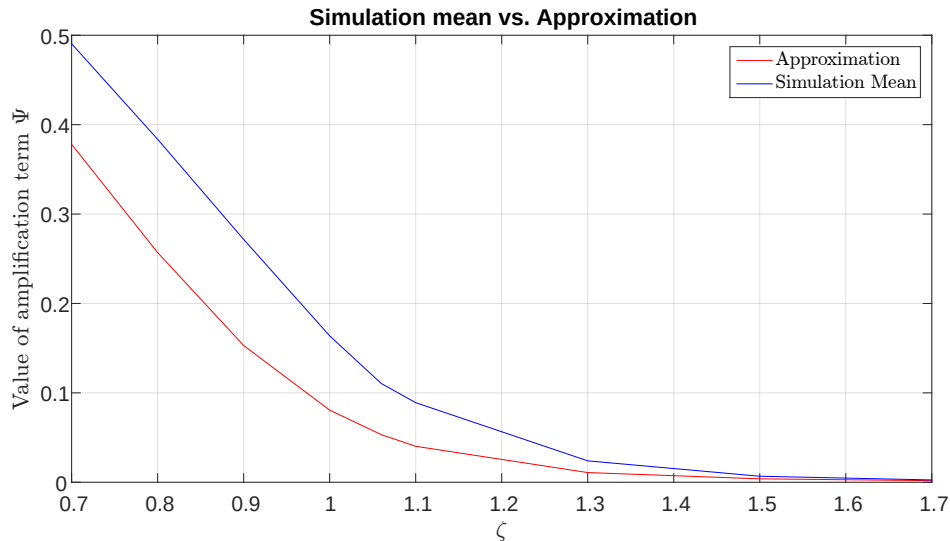
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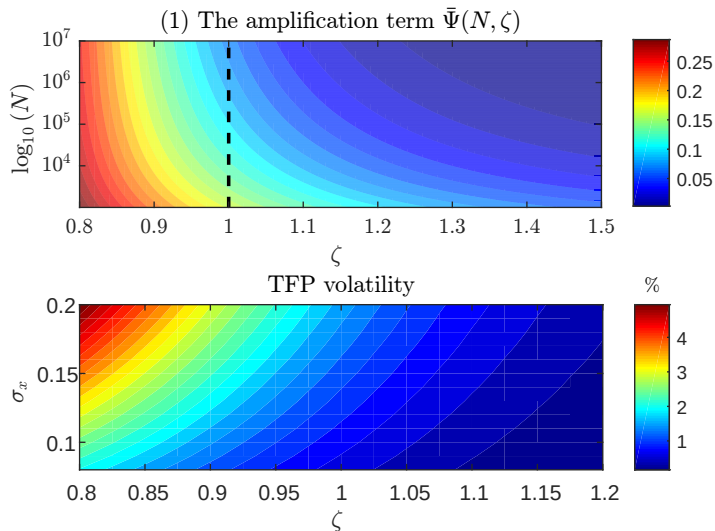
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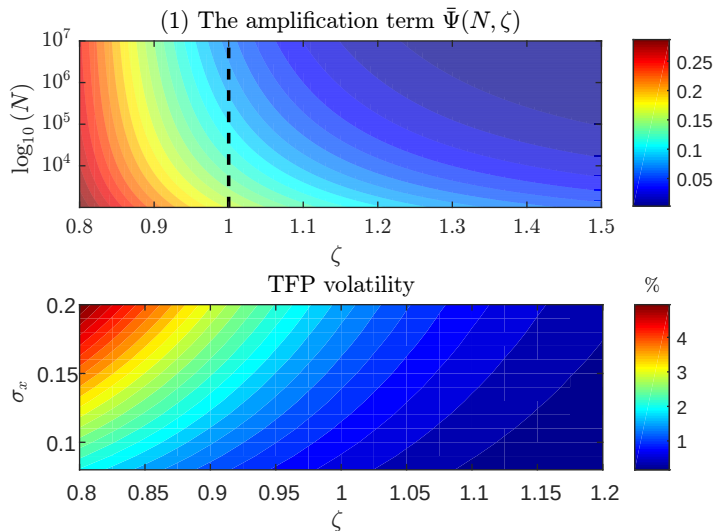
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Quantitative Validation - How Volatile are Granular Business Cycles?



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Annual TFP vol. $\approx 1\%$

**Carvalho and Grassi
(2019):**

$$\zeta = 1.1, \sigma_x = 0.08$$



25% - granularity

**Validation - my
approximation:**

$$\zeta = 1.1, \sigma_x = 0.08$$



32% - granularity

Implications

Gains From Eliminating Misallocation Ignore Business Cycle Implications

- Estimated gains from reducing misallocation are massive

$$\text{misallocation} \downarrow \Rightarrow TFP \uparrow$$

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	Efficient (CG19)	With misallocation	W/o misallocation
ζ	1.1	1.1	1
σ_{TFP}	0.32%	0.32%	0.56%

Robustness

Conclusions

- 1 There is a **tradeoff between maximizing production efficiency and fostering macro stability**
- 2 **Misallocation counterfactuals have sizable business cycle implications**
- 3 **Rising concentration is a stability concern** - similar logic to macro-prudential policy
- 4 **Size-dependent policies, employment protection and SME subsidies are ex-ante automatic stabilizers**

Thank you!

Allowing for Short-term Factor Mobility - Indeterminate Results

If we allow the planner to allocate labor ex-post [Back](#)

$$\overline{\eta}_i^d = s_{Yi}^d + \frac{\gamma}{1-\gamma} \left[\underbrace{s_{Yi}^d}_{\text{sale share}} - \underbrace{s_{Li}^d}_{\text{labor share}} \right]$$

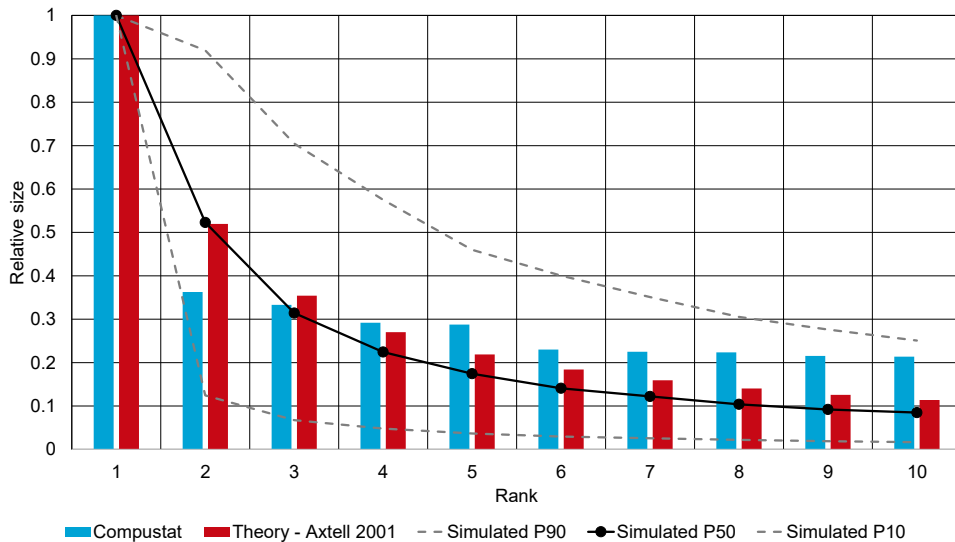
Therefore amplification is given by:

$$\Psi_d^2 = \left[1 + 2\frac{\gamma}{1-\gamma} + \left(\frac{\gamma}{1-\gamma} \right)^2 \right] \underbrace{\sum_{i=1}^N (s_{Yi}^d)^2}_{\text{Sales HHI}} + \left(\frac{\gamma}{1-\gamma} \right)^2 \underbrace{\sum_{i=1}^N (s_{Li}^d)^2}_{\text{Labor HHI}} - 2 \left[\frac{\gamma}{1-\gamma} + \left(\frac{\gamma}{1-\gamma} \right)^2 \right] \underbrace{\sum_{i=1}^N s_{Yi}^d s_{Li}^d}_{\text{Olley-Pakes term}}$$

Intuition:

- 1 First two terms decline with positive correlated distortions, but the third is ambiguous
- 2 Negative shocks to firms can generate positive effects and vice versa (BF2020)

Rank Size Relationship



Main

Stability Implications of Misallocation Counterfactuals

	(a)	(b)	(c)
	Efficient case	With misallocation	W/o misallocation
(1) Baseline $\phi = 0.09$	0.32%	0.32%	0.56%
(2) $\phi = 0.05$	0.32%	0.32%	0.44%
(3) $\phi = 0.2$	0.32%	0.32%	1.06%
(4) $\phi = 0.3$	0.32%	0.32%	1.74%
(5) Perfect mobility $\phi = 0.09$	0.32%	0.58%	0.64%
(6) Perfect mobility $\phi = 0.05$	0.32%	0.46%	0.47%
(7) Perfect mobility $\phi = 0.2$	0.32%	1.04%	1.37%
(8) Perfect mobility $\phi = 0.3$	0.32%	1.67%	2.34%

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